

Numerical Addendum: RingSusceptibility Calculator

Generated/Reviewed Documentation (Codex/Opus/Human) for v0.0.1

The calculator is a cutoff sum of closed-form ring kernels. The numerical objects are the shared SVD ring geometry, the ring-centre spatial query and filters that decide which pairs enter the sum, the McConnell form tensor with the bond axis replaced by the ring normal, and the shared $T_0/T_1/T_2$ projection. The listings below are distilled from the source: loops and bookkeeping are trimmed, but the constants, guards, inequalities, tensor entries, and output shape are the implementation's.

1 Ring sources and candidates

`conf.ring_geometries` gives the calculator one source geometry per aromatic ring. Each entry carries a centre c , a unit normal $\hat{n}$, a mean radius R , and the ordered vertices. The centre is the mean of the vertex coordinates. The normal is the least-variance direction of the mean-centred vertex cloud, with its sign fixed by the first two ordered edges. The radius is the mean centre-to-vertex distance. The full SVD plane-fit treatment is in the Haigh–Mallion numerical addendum; this calculator consumes only the fitted centre, normal, and mean radius.

```
geo.center = mean(geo.vertices);

Eigen::MatrixXd coords(geo.vertices.size(), 3);
for (size_t i = 0; i < geo.vertices.size(); ++i)
    coords.row(i) = (geo.vertices[i] - geo.center).transpose();

Eigen::JacobiSVD<Eigen::MatrixXd> svd(coords, Eigen::ComputeFullV);
geo.normal = svd.matrixV().col(2);          // least spread

Vec3 e1 = geo.vertices[1] - geo.vertices[0];
Vec3 e2 = geo.vertices[2] - geo.vertices[0];
if (geo.normal.dot(e1.cross(e2)) < 0) geo.normal = -geo.normal;

geo.radius = mean(norm(vertex - geo.center));
```

For each atom, the candidate set comes from the nanoflann tree over ring centres. The query point is the atom position and the radius is `ring_current_spatial_cutoff`, 15.0 Å in the shared TOML file. The radius query decides only which ring indices are returned. A ring not returned by the query never reaches the kernel helper; a returned ring can still be omitted by the filters below.

```
auto nearby_rings =
    spatial.RingsWithinRadius(atom_pos,
        cfg("ring_current_spatial_cutoff"));    // 15.0 Å

for (size_t ri : nearby_rings) {
    const RingGeometry& geom = conf.ring_geometries[ri];
    RingChiKernelResult kernel =
        ComputeRingChiKernel(atom_pos, geom.center, geom.normal);
    ...
}
```

2 One ring kernel

For an atom at $\mathbf{r}_i$, the helper forms $\mathbf{x} = \mathbf{r}_i - \mathbf{c}$, the displacement from ring centre to atom. Its length is $r = |\mathbf{x}|$, and $\hat{\mathbf{d}} = \mathbf{x}/r$ is the unit centre-to-atom direction. The angle used by the kernel enters only through

$$u = \cos \theta_k = \hat{\mathbf{d}} \cdot \hat{\mathbf{n}}.$$

If $r < 0.1 \text{ \AA}$, the `singularity_guard_distance`, the helper returns the zero-initialized result. In that return path the stored distance remains zero, so the later distance filter rejects the candidate instead of recording a zero-valued neighbour.

```
Vec3 x = atom_pos - ring_center;          // center -> atom
double r = x.norm();
if (r < cfg("singularity_guard_distance")) return result;

result.distance = r;
double r3 = r * r * r;
Vec3 d_hat = x / r;
double u = d_hat.dot(ring_normal);        // cos(theta_k)

result.scalar_kernel = (3.0 * u * u - 1.0) / r3;
```

The formula is the McConnell bond-anisotropy kernel with the bond axis $\hat{\mathbf{b}}$ replaced by the ring normal $\hat{\mathbf{n}}$. Under the substitution

$$\hat{\mathbf{b}} \mapsto \hat{\mathbf{n}}, \quad c = \hat{\mathbf{d}} \cdot \hat{\mathbf{b}} \mapsto u = \hat{\mathbf{d}} \cdot \hat{\mathbf{n}},$$

the ring helper has the same scalar $f = (3u^2 - 1)/r^3$, the same dipolar helper

$$K_{ab} = \frac{3\hat{d}_a\hat{d}_b - \delta_{ab}}{r^3},$$

and the same three-term tensor structure:

$$M_{ab} = \frac{9u\hat{d}_a\hat{n}_b - 3\hat{n}_a\hat{n}_b - (3\hat{d}_a\hat{d}_b - \delta_{ab})}{r^3}.$$

The helper stores K , but `RingSusceptibility` does not write it to any feature field. The per-ring neighbour fields store M , with units $\text{\AA}^{-3}$. This class does not multiply by a ring-current intensity, a Johnson–Bovey lobe offset, a susceptibility scale, or a ring-type weight.

```
for (int a = 0; a < 3; ++a)
  for (int b = 0; b < 3; ++b)
    result.dipolar_kernel(a,b) =
      (3.0*d_hat(a)*d_hat(b) - (a == b ? 1.0 : 0.0)) / r3;

for (int a = 0; a < 3; ++a) {
  for (int b = 0; b < 3; ++b) {
    double Iab = (a == b ? 1.0 : 0.0);
    result.full_tensor_over_r3(a,b) =
      (9.0*u*d_hat(a)*ring_normal(b)
       - 3.0*ring_normal(a)*ring_normal(b)
       - (3.0*d_hat(a)*d_hat(b) - Iab)) / r3;
  }
}
```

The three terms determine which tensor channels can be populated. The trace of the first term is $9u^2$, the trace of the second is -3 , and the third term is traceless, so

$$T_0(M) = \frac{1}{3} \text{tr } M = \frac{3u^2 - 1}{r^3} = f.$$

The antisymmetric part can only come from $9u \hat{\mathbf{d}} \otimes \hat{\mathbf{n}}$. It is proportional to $u(\hat{\mathbf{d}} \otimes \hat{\mathbf{n}} - \hat{\mathbf{n}} \otimes \hat{\mathbf{d}})$, so it vanishes when $\hat{\mathbf{d}}$ is parallel or antiparallel to $\hat{\mathbf{n}}$ (the atom on either end of the ring axis, $u = \pm 1$) and when $u = 0$ (the atom in the ring plane). Away from those geometries it can survive as T_1 . The symmetric traceless remainder fills T_2 .

3 Ring-frame coordinates

The shared neighbour record also carries the accepted atom in a cylindrical frame tied to the same fitted ring geometry. When RingSusceptibility creates the record, the axial coordinate is

$$z = \mathbf{x} \cdot \hat{\mathbf{n}},$$

and the in-plane radial vector is $\mathbf{x}_\perp = \mathbf{x} - z\hat{\mathbf{n}}$, with $\rho = |\mathbf{x}_\perp|$. Since $\hat{\mathbf{n}}$ is unit length, $r^2 = z^2 + \rho^2$. The kernel angle above uses the signed cosine $u = z/r$. The stored polar angle is different:

$$\theta_{\text{store}} = \text{atan2}(\rho, |z|), \quad \cos \theta_{\text{store}} = |z|/r = |u|.$$

The absolute value folds the two faces of the ring together. The sign of z still records which face the atom lies on, with the ring normal's code-defined orientation, but θ_{store} alone cannot distinguish (z, ρ) from $(-z, \rho)$. Its range is $[0, \pi/2]$, from on-axis to in-plane.

```
Vec3 atom_offset = atom_pos - geom.center;
double z = atom_offset.dot(geom.normal);
Vec3 d_plane = atom_offset - z * geom.normal;
double rho = d_plane.norm();
double theta = std::atan2(rho, std::abs(z));

new_neighbour.z = z;
new_neighbour.rho = rho;
new_neighbour.theta = theta;
```

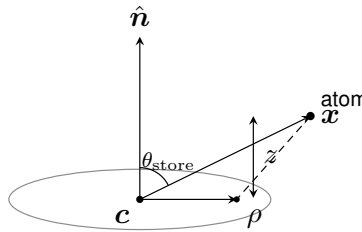


Figure 1: The neighbour coordinates split the centre-to-atom offset into axial z and radial ρ . The stored angle uses $|z|$, so points on the two faces have the same θ_{store} when their $|z|$ and ρ match.

4 Filters and omissions

The per-atom path calls the kernel helper before it applies the filter set, but a rejected pair changes no accumulators. It does not create or update a ring-neighbour record, does not add χ_{scalar} , and does not add M to the atom total.

```
filters.Add(std::make_unique<MinDistanceFilter>());
filters.Add(std::make_unique<DipolarNearFieldFilter>());
filters.Add(std::make_unique<RingBondedExclusionFilter>(protein));
```

```

ctx.distance = kernel.distance;
ctx.source_extent = 2.0 * geom.radius;           // ring diameter
ctx.atom_index = ai;
ctx.source_ring_index = ri;
if (!filters.AcceptAll(ctx)) continue;

```

MinDistanceFilter requires

$$r \geq \text{singularity_guard_distance} = 0.1 \text{ \AA}.$$

For a closer point the helper has already returned a zero tensor and left `kernel.distance` at zero; the filter turns that guarded helper result into an omitted pair. The omitted term is not replaced by a clamped $r = 0.1 \text{ \AA}$ contribution.

DipolarNearFieldFilter uses the ring diameter $2R$ as `source_extent`. With the configured ratio 0.5, the atom path accepts only

$$r > 0.5(2R) = R.$$

This is a centre-distance test against the mean-radius sphere. It is not a point-in-polygon test on the ring plane, and it does not use ρ . A candidate at or inside that boundary leaves the sum.

RingBondedExclusionFilter is topological. At construction it walks each ring vertex and all bonds incident on that vertex, then stores a per-ring set containing the vertices and their covalently bonded neighbours. If the target atom is in the set for the source ring, the pair is omitted even when the distance filters would accept it.

The grid-sampling path uses the same kernel and the same configured distances, but not the atom filter set. It loops over all aromatic rings, skips $r < 0.1 \text{ \AA}$, skips $r < R$, and skips $r > 15.0 \text{ \AA}$. It therefore keeps equality at the near-field boundary $r = R$, while the per-atom filter path rejects equality. It has no bonded exclusion because a free grid point has no atom index or covalent topology.

```

double r = (point - geom.center).norm();
if (r < cfg("singularity_guard_distance")) continue;
if (r < geom.radius) continue;
if (r > cfg("ring_current_spatial_cutoff")) continue;

auto kernel = ComputeRingChiKernel(point, geom.center, geom.normal);
M_total += kernel.full_tensor_over_r3;

```

`near_zero_vector_norm_threshold` is present in the shared parameter file with value 10^{-10} , but this calculator does not read it.

5 Accumulation and projection

Accepted per-ring tensors are added as raw 3×3 matrices:

$$M_i = \sum_{R \in \mathcal{A}_i} M^{(iR)}.$$

Both the per-ring neighbour tensor and the per-atom total go directly to `SphericalTensor::Decompose`. There is no symmetrization $\frac{1}{2}(M + M^T)$ and no trace removal before the call.

```

neighbour->chi_tensor = kernel.full_tensor_over_r3;
neighbour->chi_spherical =
    SphericalTensor::Decompose(kernel.full_tensor_over_r3);
neighbour->chi_scalar = kernel.scalar_kernel;

```

```

M_total += kernel.full_tensor_over_r3;

atom.ringchi_shielding_contribution =
    SphericalTensor::Decompose(M_total);

```

This is the full-tensor path from the McConnell calculator, with no category T_2 path beside it. McConnell reports some bond-category tensors after symmetrizing and subtracting the trace, and also reports the raw full M_{total} decomposition. RingSusceptibility uses only the raw full decomposition. The consequence is that the antisymmetric part of the first dyad is retained as nonzero T_1 whenever the accepted geometry leaves one.

The shared projection is closed-form arithmetic on a 3×3 tensor X . The trace gives T_0 , the antisymmetric part gives the Cartesian Levi-Civita dual T_1 , and the symmetric traceless part gives the five stored T_2 numbers.

```

st.T0 = sigma.trace() / 3.0;

st.T1[0] = 0.5*(sigma(1,2) - sigma(2,1));
st.T1[1] = 0.5*(sigma(2,0) - sigma(0,2));
st.T1[2] = 0.5*(sigma(0,1) - sigma(1,0));

double Sxx = sigma(0,0) - st.T0;
double Syy = sigma(1,1) - st.T0;
double Szz = sigma(2,2) - st.T0;
double Sxy = 0.5*(sigma(0,1) + sigma(1,0));
double Sxz = 0.5*(sigma(0,2) + sigma(2,0));
double Syz = 0.5*(sigma(1,2) + sigma(2,1));

st.T2[0] = std::sqrt(2.0) * Sxy;
st.T2[1] = std::sqrt(2.0) * Syz;
st.T2[2] = std::sqrt(3.0/2.0) * Szz;
st.T2[3] = std::sqrt(2.0) * Sxz;
st.T2[4] = (Sxx - Syy) / std::sqrt(2.0);

```

The T_2 constants make the five-number packing isometric. An off-diagonal entry such as S_{xy} appears twice in the symmetric matrix, as S_{xy} and S_{yx} , so it contributes $2S_{xy}^2$ to the Frobenius sum. The stored $\sqrt{2} S_{xy}$ squares to the same amount. The diagonal entries have only two degrees of freedom after $S_{xx} + S_{yy} + S_{zz} = 0$; the $\sqrt{3/2} S_{zz}$ and $(S_{xx} - S_{yy})/\sqrt{2}$ entries carry those two degrees so their squares add back to the diagonal Frobenius contribution.

ringchi_shielding.npy has shape $N \times 9$, one row per atom, in the shared packed order

$$[T_0, T_{1x}, T_{1y}, T_{1z}, T_{2,-2}, T_{2,-1}, T_{2,0}, T_{2,+1}, T_{2,+2}].$$

6 Glossary

Dyadic product. A grid where one vector labels the rows and the other labels the columns, every cell holding the product of its two labels. In this calculator, $\hat{\mathbf{d}} \otimes \hat{\mathbf{n}}$ has entry $\hat{d}_a \hat{n}_b$, so swapping the indices can change the value. Formally, $(\mathbf{p} \otimes \mathbf{q})_{ab} = p_a q_b$.

Dipolar helper K . The bar-magnet field shape laid along the centre-to-atom line — strongest and signed down that axis, weaker and reversed to the side (a symmetric traceless tensor), assembled before the full ring tensor but not emitted. It has eigenvalue $+2/r^3$ along $\hat{\mathbf{d}}$ and $-1/r^3$ in each transverse direction. Formally, $K_{ab} = (3\hat{d}_a \hat{d}_b - \delta_{ab})/r^3$.

Near-field exclusion. A bubble drawn around the ring, sized to the real loop the point kernel stands in for (a centre-distance exclusion sphere whose radius is the mean ring radius). For a ring with mean radius R , it omits the candidate unless $r > R$. Formally, the filter is $r > \rho_{\text{nf}} E$, with $E = 2R$ and $\rho_{\text{nf}} = 0.5$.

Spherical tensor split. A 3×3 tensor holds three kinds of content that rotate without mixing — an orientation-free size (the scalar trace, T_0), an axial twist from its antisymmetric part (the three T_1 components), and a five-number traceless shape for the rest of the direction-dependence (the five T_2 components). RingSusceptibility feeds the raw asymmetric McConnell form tensor into that split, so all nine stored numbers can be populated. Formally, it is the closed-form split of a 3×3 tensor's nine components into dimensions $1 + 3 + 5$, with the T_2 basis normalized to preserve the Frobenius norm.